IMPERIAL

# NanoStat: A Handheld Potentiostat for Cortisol Aptasensor Readout

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NanoStat integrates square-wave voltammetry (SWV) acquisition, BLE/browser data export, and a guarded low-leakage PCB layout for portable aptasensor readout, as shown in Figure 1.

## 1. Background & Objective

- Cortisol is a stress-related biomarker that can be measured non-invasively in saliva [1].
- This project targets square-wave-voltammetry (SWV) peak-current changes from a methylene-blue (MB)-labelled cortisol aptasensor [2].
- NanoStat was developed as an 11 × 6 × 2 cm portable potentiostat for tens-to-hundreds-of-nA electrochemical readout, with BLE connectivity and an estimated one-week battery lifetime from a 190 mAh Li-Po battery.
- The platform was validated in Phosphate-Buffered Saline (PBS) against an Autolab reference system to assess whether the portable reader can reproduce concentration-dependent SWV responses.

## 2. System Architecture

Figure 1 illustrates the NanoStat system architecture.

- AD5941 Analogue Front End (AFE) for electrode biasing, transimpedance-amplifier (TIA) current readout, ADC sampling and sequenced SWV [3].
- nRF52840 module for BLE communication, SPI control and user workflow management.
- nPM1300 PMIC for USB/battery operation and controlled AFE power isolation.
- OLED Display, push buttons and a flash memory.

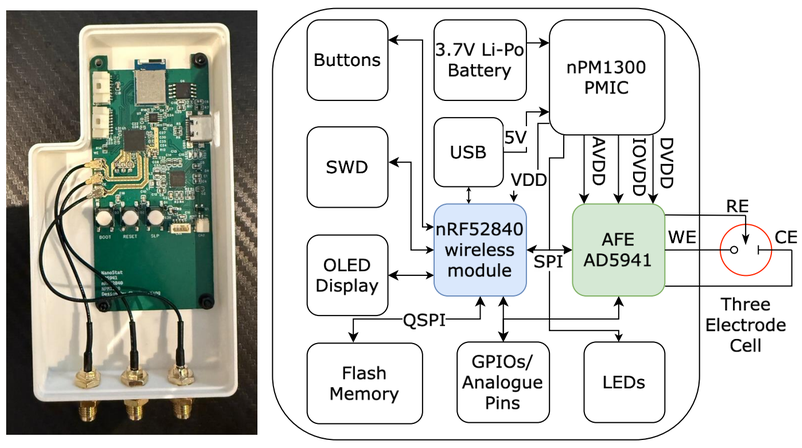



Fig. 1. NanoStat prototype (battery and OLED not shown) and system architecture.

## 3. AD5941 SWV Implementation

Figure 2 shows how SWV was implemented:

- PBS SWV was generated using the AD5941 LPDAC, switch matrix, HSTIA, ADC, FIFO and sequencer [3].
- The sequencer updates VBIAS/VZERO during the scan while sampled current data are temporarily stored in FIFO before being read by the nRF52840.

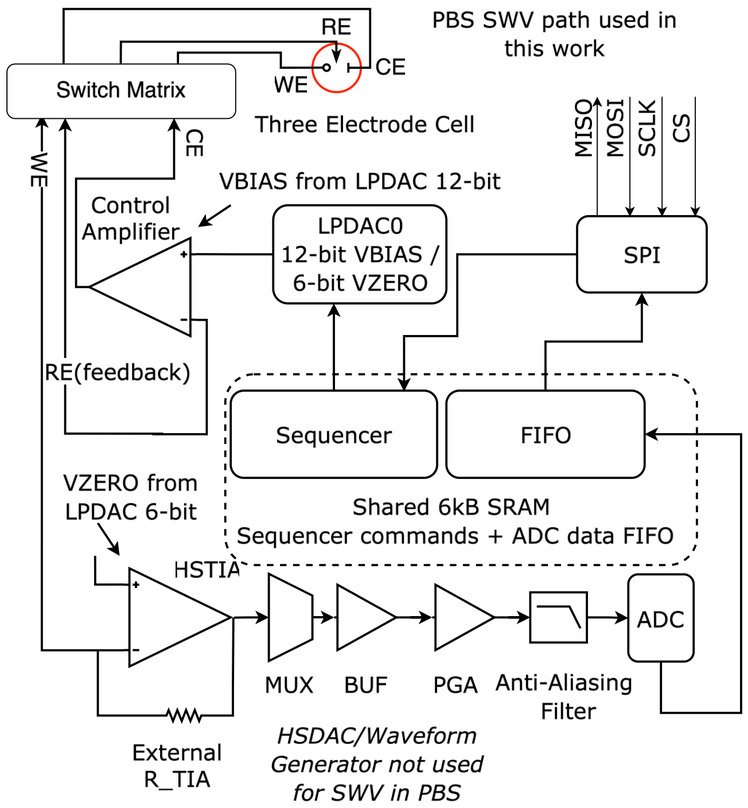



Fig. 2. AD5941 signal path used for PBS SWV acquisition.

[1] A. N. Naeem, S. Guldin, and S. S. Ghoreishizadeh, "Electrochemical sensors for cortisol: A review," Mar. 2024
[2] S. S. Ghoreishizadeh, A. N. Naeem, et al., "Development of an electrochemical aptasensor for rapid measurement of cortisol in saliva," in review.
[3] Analog Devices, "AD5940/AD5941: High precision, impedance, and electrochemical front end," Data Sheet, Rev. G, 2026.
[4] V. Mirčeski, S. Komorsky-Lovrić, and M. Lovrić, Square-Wave Voltammetry: Theory and Application. Berlin, Germany: Springer, 2007.

## 4. PCB Layout for Low-Current Readout

- Low-current electrochemical measurements are sensitive to leakage current and parasitic coupling.
- As shown in Figure 3, driven guard structures were placed around high-impedance summing nodes and sensitive electrode traces to reduce leakage-related error and improve low-current readout robustness.

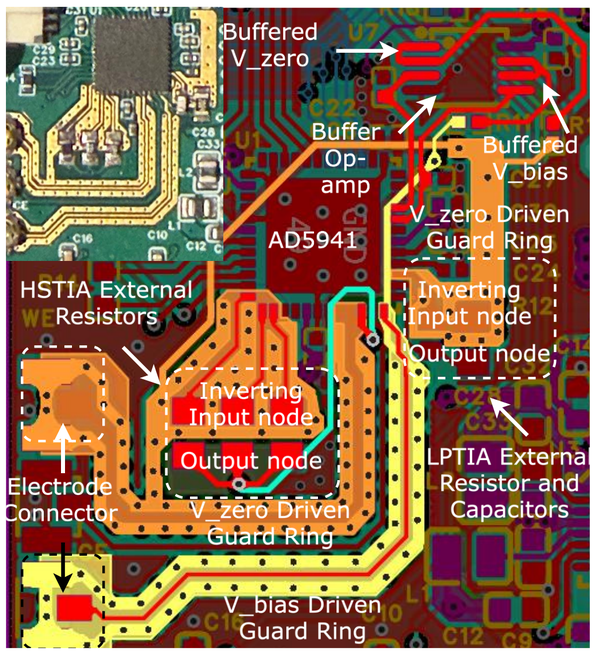



Fig. 3. Guarded PCB layout around the low leakage current readout path.

## 5. Measurement and Signal Processing Working Flow

Figure 4 illustrate the NanoStat measurement workflow. After each scan, the electrode paths are switched to a high-Z state to protect the sensor.

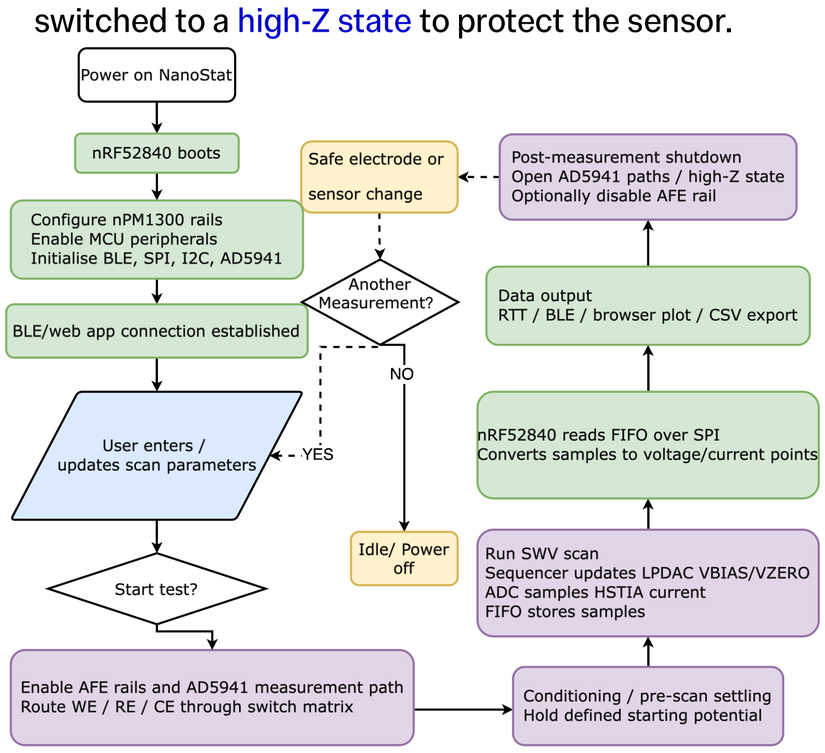



Fig. 4. NanoStat measurement workflow

Figure 5 shows the offline peak-extraction workflow:

- NanoStat streams raw SWV points, while the browser/Python analysis tool applies SG5 smoothing, two-shoulder baseline subtraction, and fixed-window MB peak detection.
- The peak current is defined as the maximum baseline-subtracted current in the -300 to -200 mV window, using baseline shoulders at -400 to -330 mV and -170 to -100 mV.
- The same workflow is applied to both NanoStat and Autolab data.

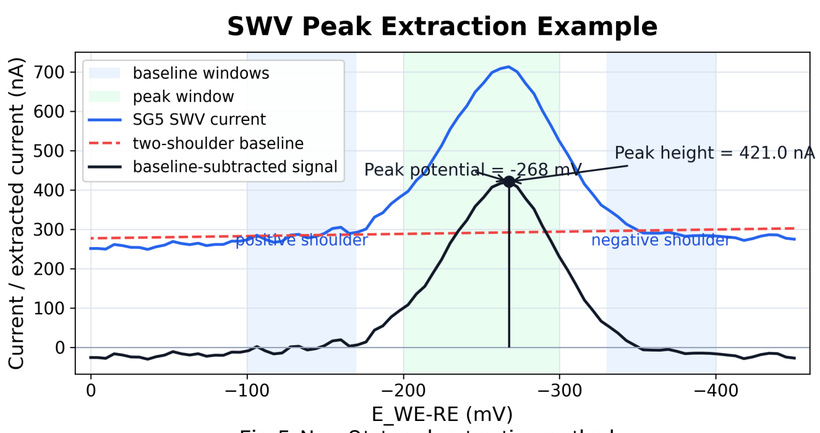



Fig. 5. NanoStat peak extraction method.

## 6. PBS Validation: NanoStat vs Autolab

### Setup & Method

Table 1 summarises the SWV parameters, and Figure 6 shows the experimental setup and programmed SWV profiles.

Note: NanoStat was powered from an unplugged MacBook USB-C port rather than the battery during PBS tests.

- USB power was disconnected after each sensor scan to guarantee a hardware-level cell-off state during electrode changes.
- This reduced the risk of unintended aptasensor polarisation before firmware high-impedance shutdown was validated, at the cost of possible USB-ground-coupled noise.

| Parameter | NanoStat | Autolab/NOVA |
|---|---|---|
| Scan range | From -0.45 V to 0 V vs RE | From -0.45 V to 0 V vs RE |
| Frequency | 120 Hz | 120 Hz |
| Modulation amplitude | ~35 mV | ~35 mV |
| Step height | 5.37 mV | 1.07 mV |
| RTIA | 330 kΩ external HSTIA RTIA | Autolab internal current range |
| CTIA | 4 pF | Autolab internal bandwidth / filtering setting |
| Conditioning / quiet time | -0.45 V held for 3 s before SWV | 0 V conditioning for 5 s, then SWV scan |

Tab.1. SWV parameters of NanoStat and Autolab

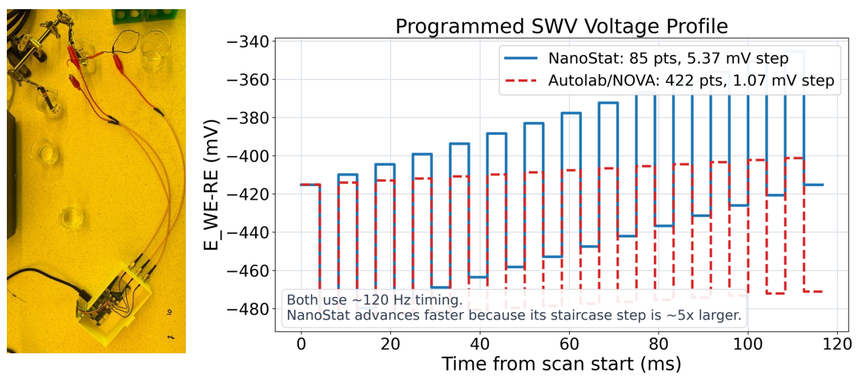



Fig. 6. NanoStat experiment setup and SWV voltage profile with comparison of Autolab reference.

### Result

NanoStat and Autolab measured the same regenerated sensors in PBS, and the concentration responses are shown in Figure 7:

- Both systems produced clear methylene-blue SWV peaks.
- Autolab traces were smoother and showed lower blank variation and produced a clearer monotonic response from 1 to 1000 ng/mL.
- NanoStat followed the same high-concentration trend, but with weaker low-concentration separation, resulting in a non-monotonic trend at low concentrations.

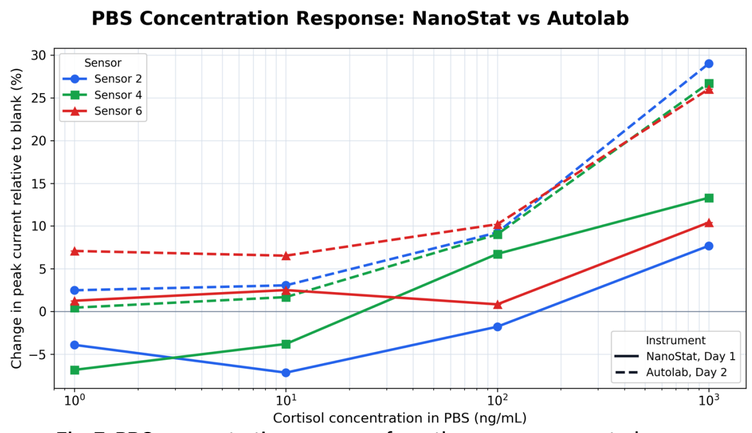



Fig. 7. PBS concentration response from the same regenerated sensors measured by NanoStat and Autolab.

## 7. Discussion

The current PBS concentration response shows that the portable readout chain works properly but with relatively low repeatability compared with the Autolab reference. This is mainly due to:

- The 5.37 mV NanoStat voltage step reduced peak resolution and filtering effectiveness, resulting in high blank variance and weak trend at low concentrations.
- The USB-C powering may introduce extra blank noise compared with Li-Po battery operation.
- The actual waveform has not been validated by oscilloscope, leaving uncertainty in signal quality.

## 8. Future Work

- Reduce step height to 1 mV by segmented sequencing.
- Use battery operation to obtain lower noise.
- Repeat PBS calibration using freshly prepared sensors and a same-day NanoStat-Autolab comparison.
- Introduce test strip electrodes for portable cortisol trend monitoring in human subjects.